

CU-423

RoboLink: Build and Drive Your Smartphone-Controlled Robot

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04-May-2026

Day 1:

Welcome to

Electronics & Electricity

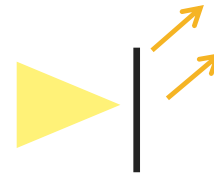
Activity

- Introduce each other

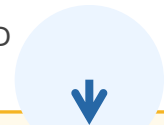


Goals

1. Explain what electricity is, then voltage, current, and resistance.
2. Read and interpret resistor colour codes
3. Explain LEDs and breadboards
4. Use Ohm's law to predict current in a simple LED circuit
5. Use KVL as the loop rule for voltage
6. Combine resistors in series and in parallel
7. Connect the ideas to LEDs and variable resistors



LED





GONNA BE GREAT, LET'S DO IT

Before We Power Anything

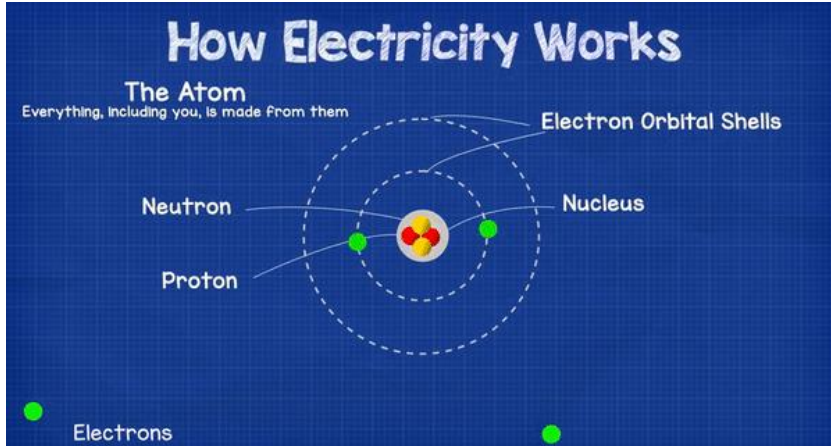
Lab habits for low-voltage robot circuits

1. Use only the batteries, USB power, and supplies provided in the lab.
2. Check the plus and minus sides before connecting the power.
3. If a part gets hot, smells odd, or the circuit acts strangely: disconnect power.
4. Never connect course parts directly to a wall outlet.



**Power OFF
before
rewiring**

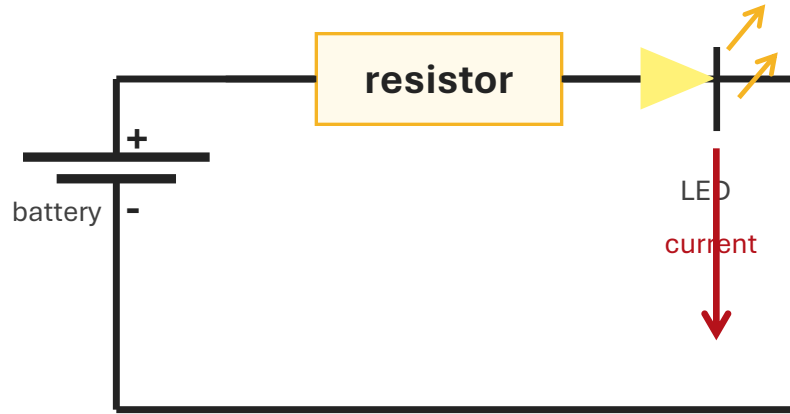
What Is Electricity?



1. Everything is made of atoms.
2. Atoms contain tiny charged particles
3. Electricity is energy carried by electric charges
4. In wires, moving charges can carry energy to a part

The main idea of a Circuit is to guide charge so it does useful work

A Circuit Is a Closed Loop

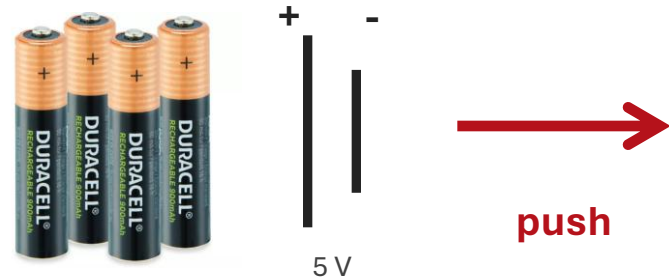
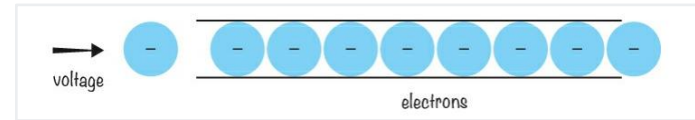


1. A source gives electrical energy
2. Wires make a path
3. A load uses energy: LED, sensor, or buzzer
4. Current flows only when the loop is closed

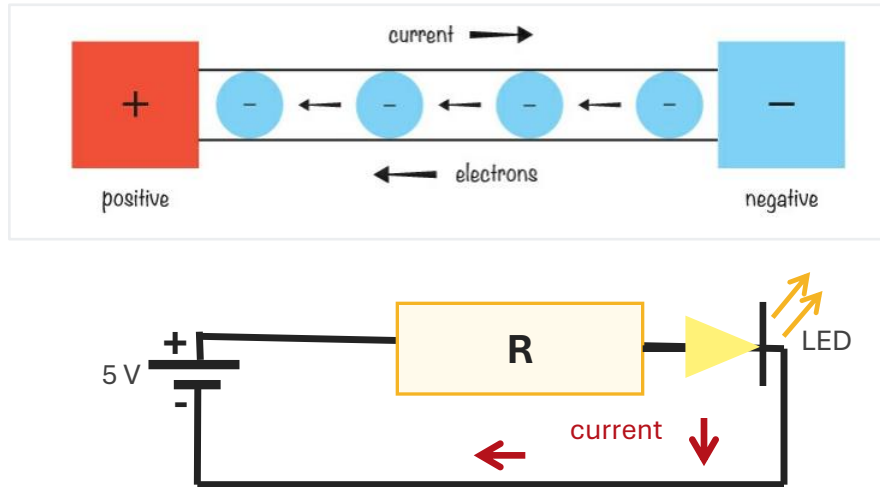
No closed loop = no current

Voltage: The Electrical Push

1. Voltage is how much electrical push a source can give.
2. Unit: volt, written V
3. A bigger voltage can push harder through the same circuit.
4. A battery has a plus side and a minus side.



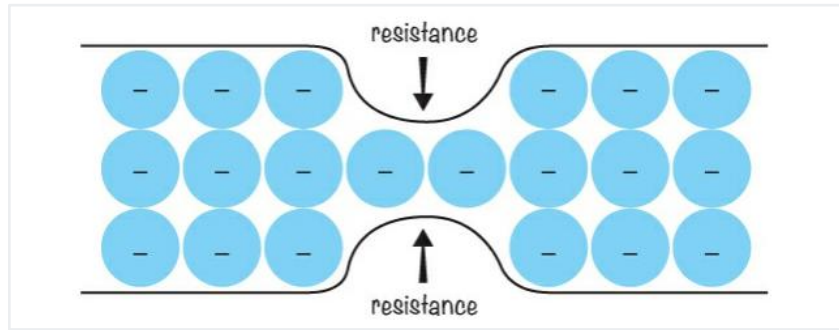
Current: The Electrical Flow



1. Current is moving electric charge
2. Unit: ampere, written A
3. Small circuits often use milliamps: mA
4. $1000 \text{ mA} = 1 \text{ A}$

**Too much current can
damage parts**

Resistance: The Current Controller



1. Resistance makes it harder for current to flow
2. Unit: ohm
3. Resistors protect LEDs and set signal levels
4. More resistance means less current when voltage is the same

Resistors turn math into safer circuits

Water Analogy

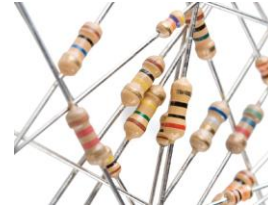
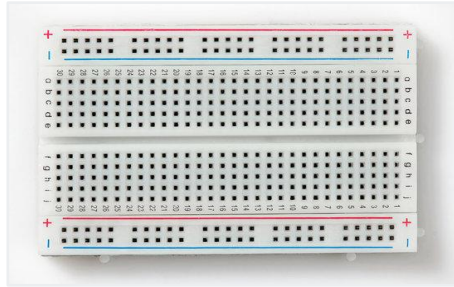
Water system

1. Pump gives pressure
2. Water flow moves through pipes
3. Narrow pipe slows flow
4. Tank stores water

Electric circuit

1. Battery gives voltage
2. Current moves through wires
3. Resistance slows current
4. Capacitor stores charge

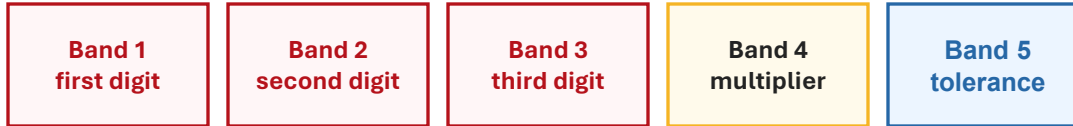
Parts We Will Use Today



1. Breadboard: temporary connections
2. Wires: carry current
3. LED: lights up in one direction
4. Resistor: limits current
5. Variable resistor: adjustable resistance

The same rules apply before and after the circuit is on the breadboard

How to Read a Resistor Colour Code



Black	0
Red	2
Yellow	4
Blue	6
Grey	8

Brown	1
Orange	3
Green	5
Violet	7
White	9

5-band: first 3 bands are digits; band 4 is multiplier; band 5 is tolerance

Colour Code Example

Red, red, black, black, gold



$$220 \times 1 = 220 \text{ ohm}$$

Red = 2

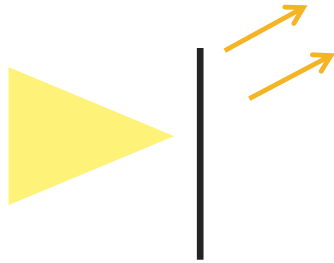
Red = 2

Black = 0

Black = x1

Gold = +/- 5%

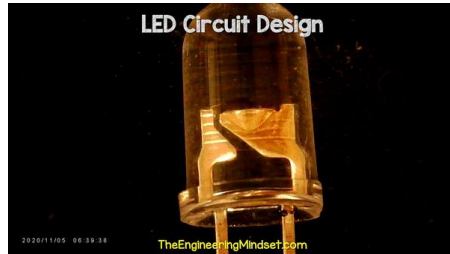
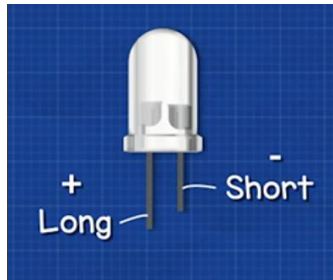
LEDs: Light-Emitting Diodes



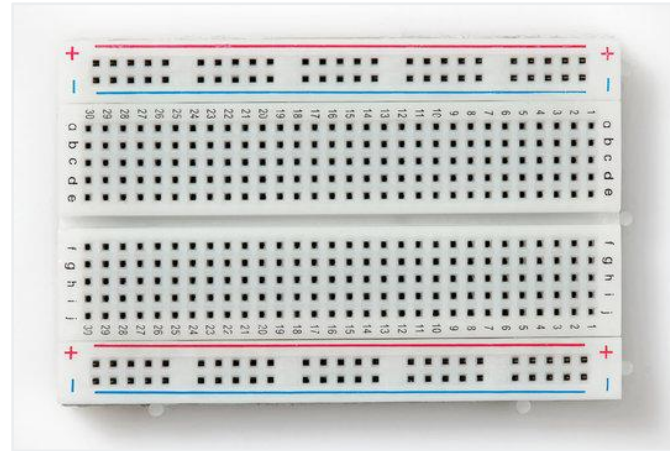
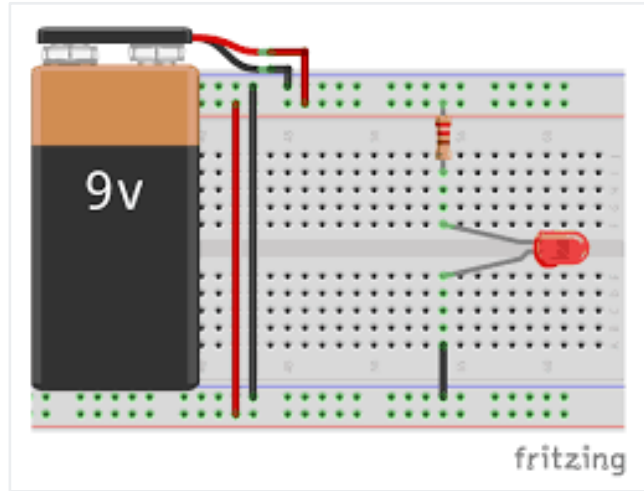
LED symbol

1. An LED lights when current flows the correct direction
2. Long leg usually connects toward plus
3. Short leg or flat side usually connects toward minus
4. Always use a resistor to limit current

Wrong direction: usually no light, and will fault the LED.



Breadboards: Connections Without Solder



1. Rows connect inside
2. Power rails run along the side
3. The centre gap separates two sides
4. Move one wire at a time

Breadboards are for building and testing circuits before permanent wiring

Variable Resistors



turn knob =
change R

1. A variable resistor lets us change resistance
2. A potentiometer is a common variable resistor
3. Turning the knob moves a contact inside
4. It can control LED brightness or create an input value

Turn knob -> change R -> change
current



Ohm's Law: One Useful Relationship

$$V = I \times R$$

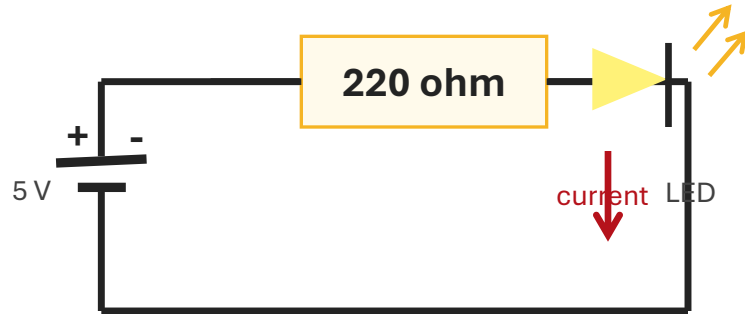
1. V is voltage in volts
2. I is current in amps
3. R is resistance in ohms

$$I = V / R$$

Use the voltage across the resistor when calculating LED current

Example: Predict LED Current

5 V source, red LED about 2 V, 220 ohm resistor



$$\text{Resistor voltage} = 5 \text{ V} - 2 \text{ V}$$

$$\text{Resistor voltage} = 3 \text{ V}$$

$$I = 3 \text{ V} / 220 \text{ ohm}$$

$$I = 0.014 \text{ A} = 14 \text{ mA}$$

Quick Check: LED Circuit

Question

A 5 V source powers an LED that drops 2 V.
The resistor is 100 ohm.
What current flows?

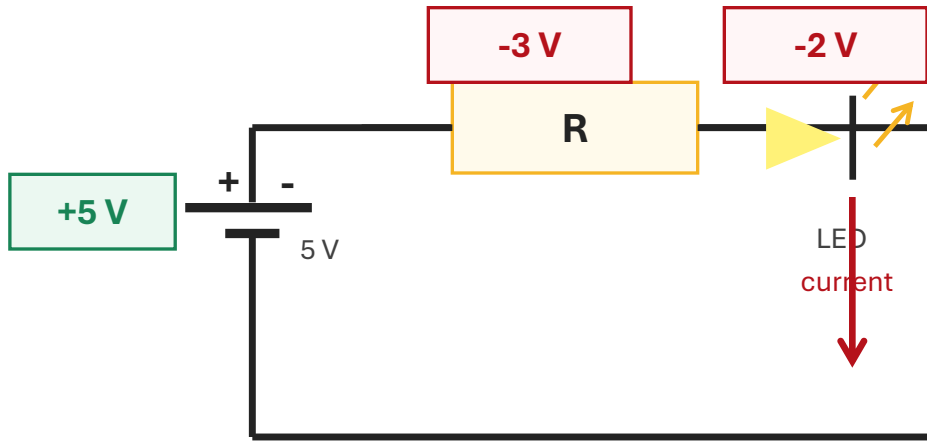
$$I = V / R$$

$$I = 3 / 100$$

$$I = 0.03 \text{ A}$$

$$I = 30 \text{ mA}$$

KVL: The Voltage Loop Rule



1. KVL means Kirchhoff's Voltage Law
2. Around one closed loop, voltage rises and drops balance
3. Source gives a rise
4. Parts take voltage drops

$$+5 \text{ V} - 3 \text{ V} - 2 \text{ V} = 0$$

KVL Example: One LED Loop

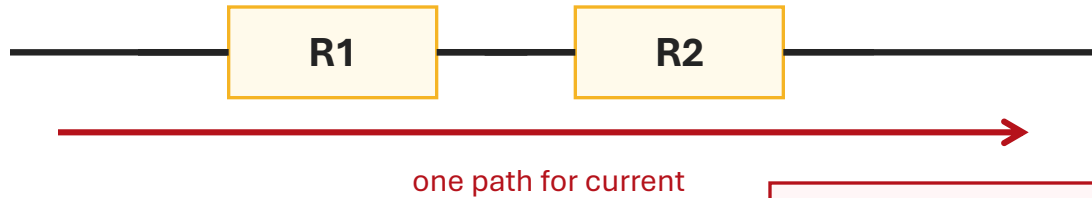
Battery voltage is shared by parts in the same loop



$$+5\text{ V} - 3\text{ V} - 2\text{ V} = 0$$

The loop balances: the source's push is used by the parts.

Resistors in Series

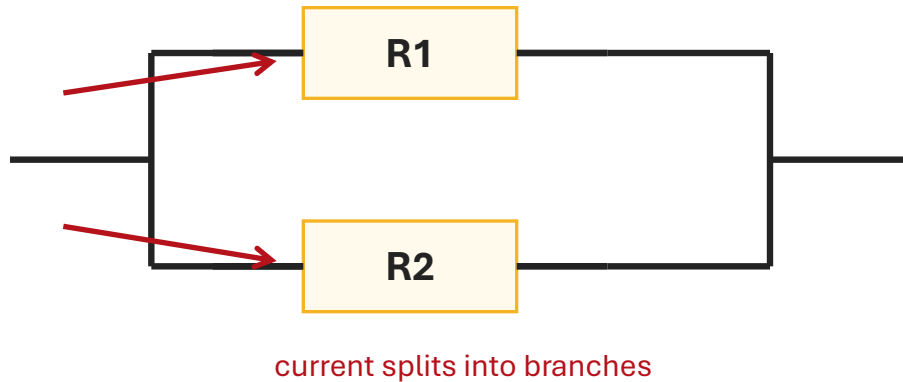


1. Current has one path
2. Total resistance gets bigger
3. Add the resistors

$$R_{\text{total}} = R1 + R2$$

$$100 + 220 = 320 \text{ ohm}$$

Resistors in Parallel



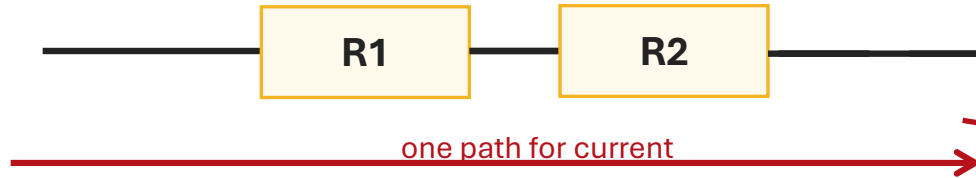
1. Current has more than one path
2. Total resistance gets smaller
3. For two equal resistors, total is half

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$100 \text{ parallel } 100 = 50 \text{ ohm}$$

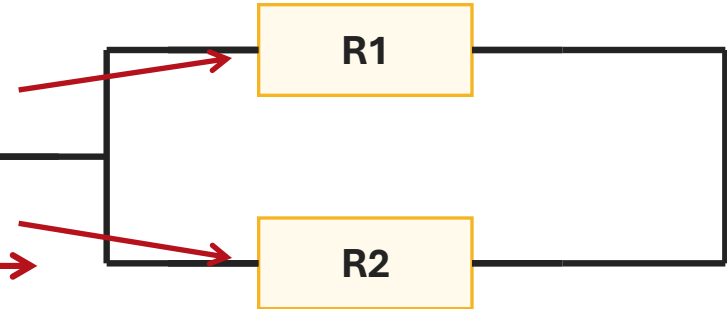
Series vs. Parallel

Series



One path; resistances add

Parallel



Many paths; total gets smaller

Mini Design Challenge

Use the rules before connecting the LED

You have:
5 V source
Red LED drops 2 V
Target current: about 15
mA

Which resistor is closest?

100

220

1000

Discuss first. Then calculate with $I = V / R$.

Vocabulary Recap

Student-friendly meaning	Symbol	Word
Energy carried by electric charges	charge	Electricity
Electrical push from a source	V	Voltage
Flow of electric charge	I	Current
How strongly a part limits current	R	Resistance
How much charge a capacitor can store	C	Capacitance
Voltage rises and drops balance around a loop	loop	KVL

Exit Ticket

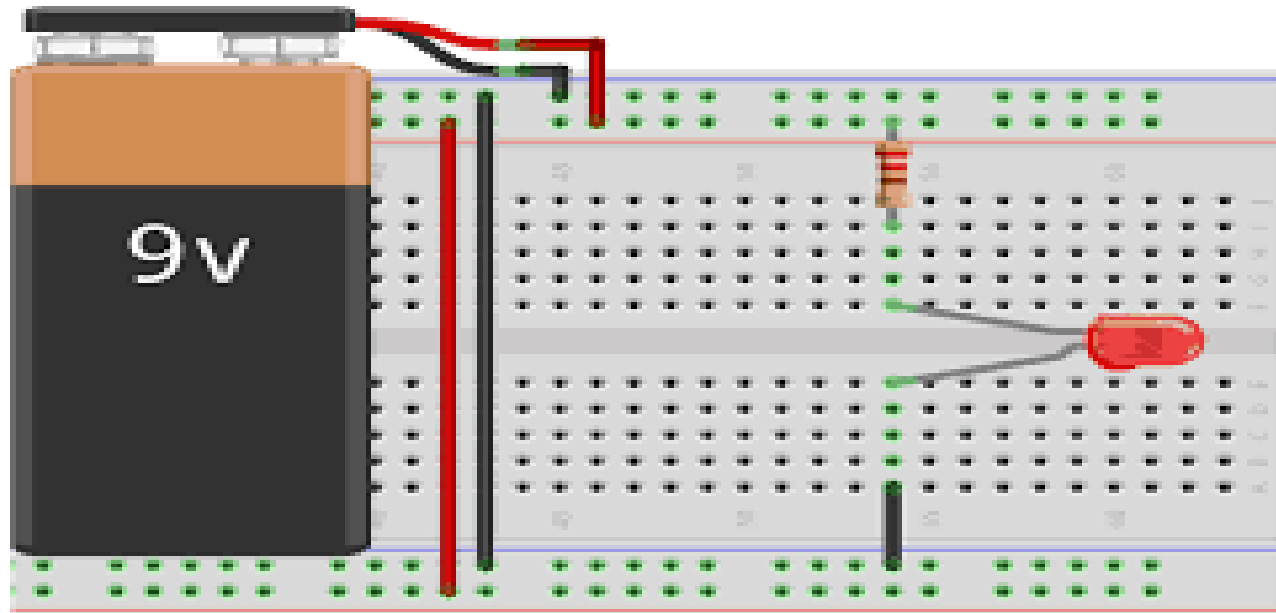
1. What does electricity mean in one sentence?
2. What does a resistor do in an LED circuit?
3. How do you read the first three colour bands on a resistor?
4. In one loop, what does KVL say about voltage rises and drops?

Next: build and test the first LED circuits

Questions?

Carleton
University





fritzing

$R = 100$
Ohms

